

Landmark Detection, Region Convolution Neural Network

R-CNN Fast R-CNN Faster R-CNN

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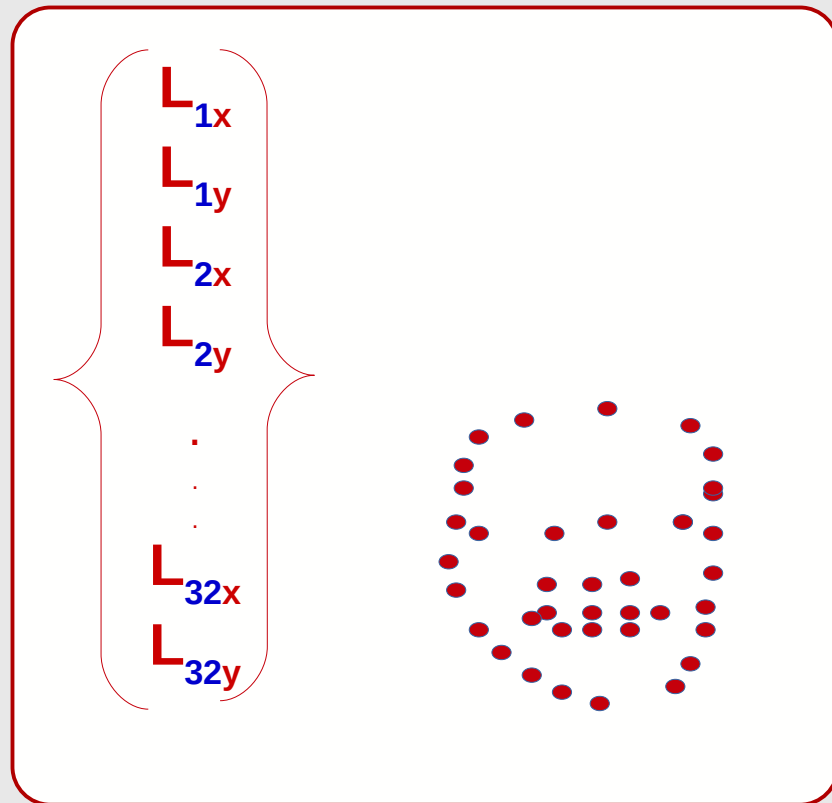
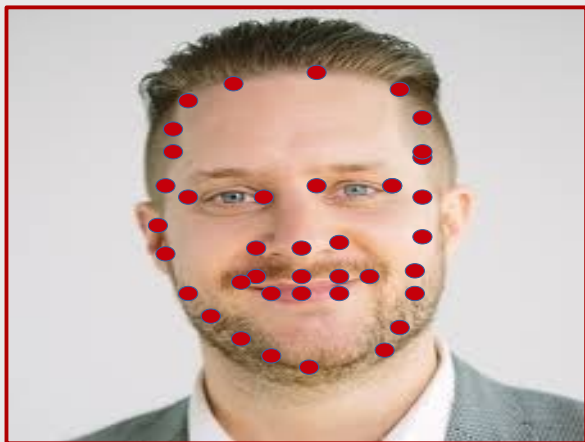
جامعة العلوم والتكنولوجيا
University of Science and Technology

Landmark Detection and Pose Estimation

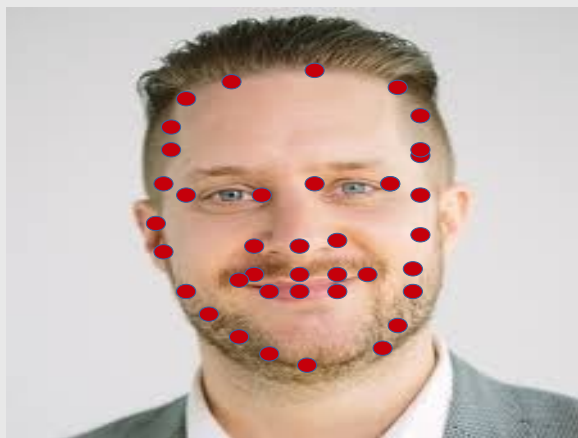
Landmark Detection



32 Landmarks
Of Face



Landmark Detection



Is it a Face
or NOT

32 Landmarks
Of Face

P_c

L_{1x}

L_{1y}

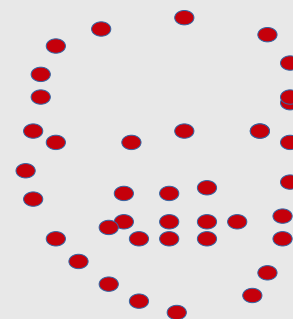
L_{2x}

L_{2y}

$\vdots$

L_{32x}

L_{32y}

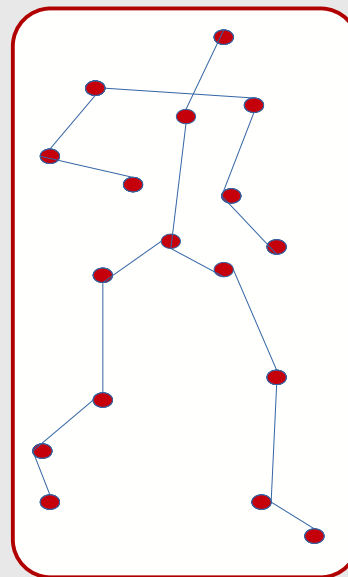


Landmark Detection



Is it a Person
or NOT

17 Landmarks
Of Pose



P_c
 L_{1x}
 L_{1y}
 L_{2x}
 L_{2y}
.
.
.
 L_{17x}
 L_{17y}

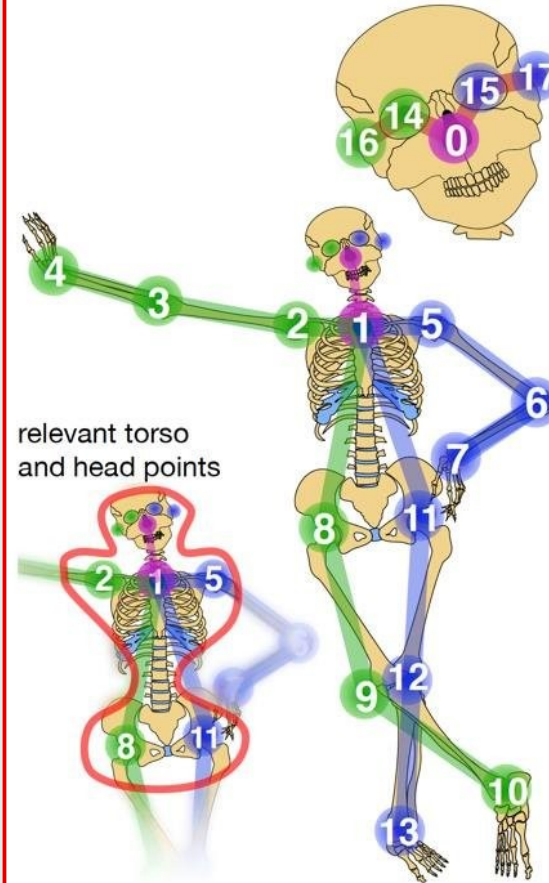
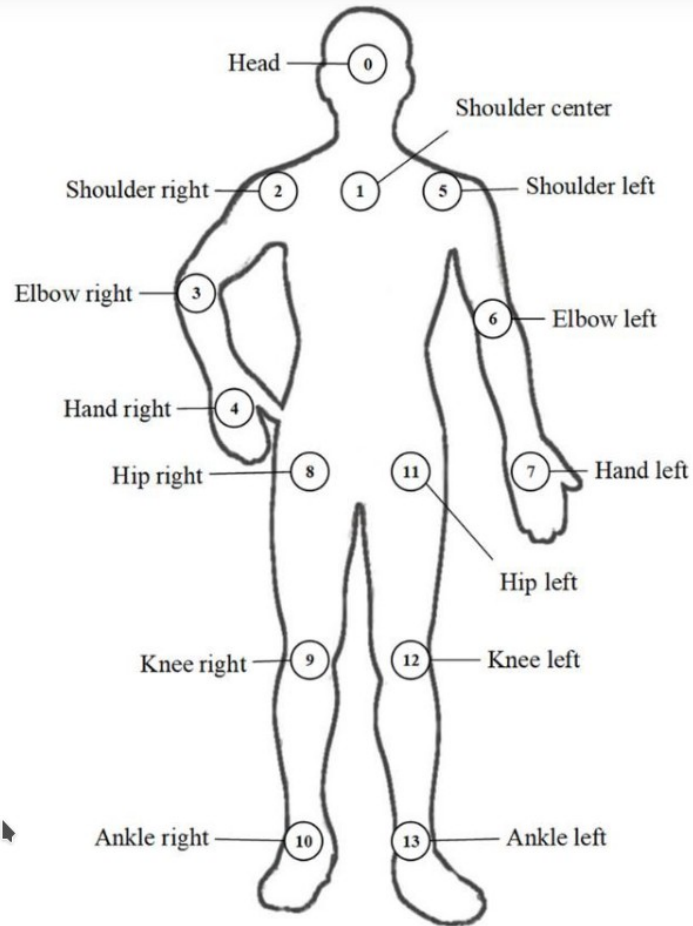
Applications of Landmark Detection



- Emotion Detection
- Augmented Reality Applications
- Adding Graphical Objects to Faces (e.g. Hat)
- Pose Detection
-
-

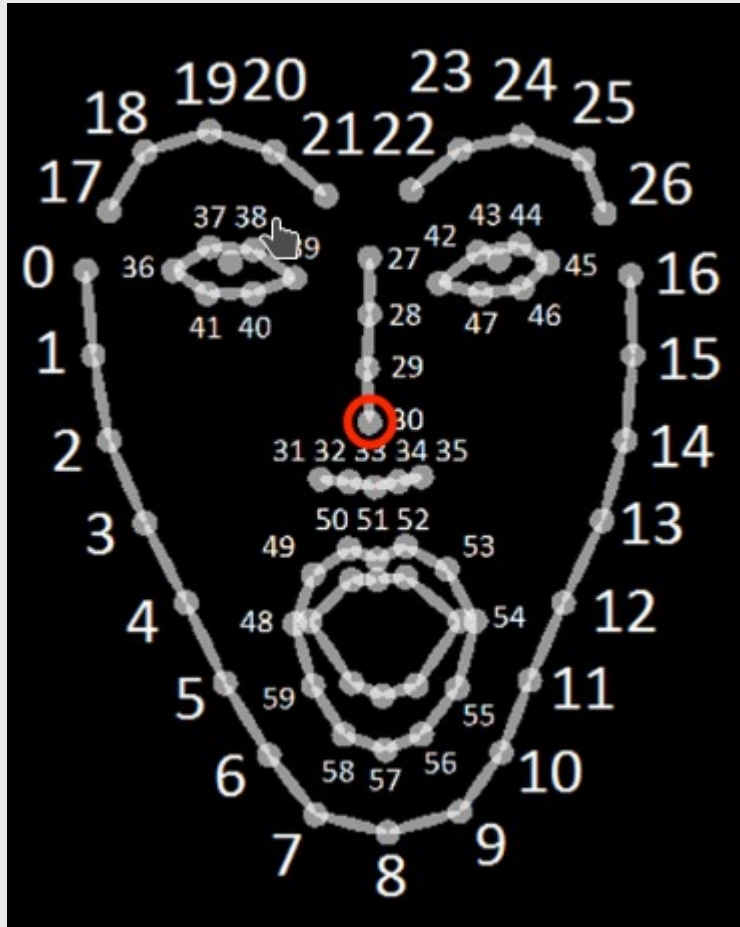
Human Pose Keypoints

7



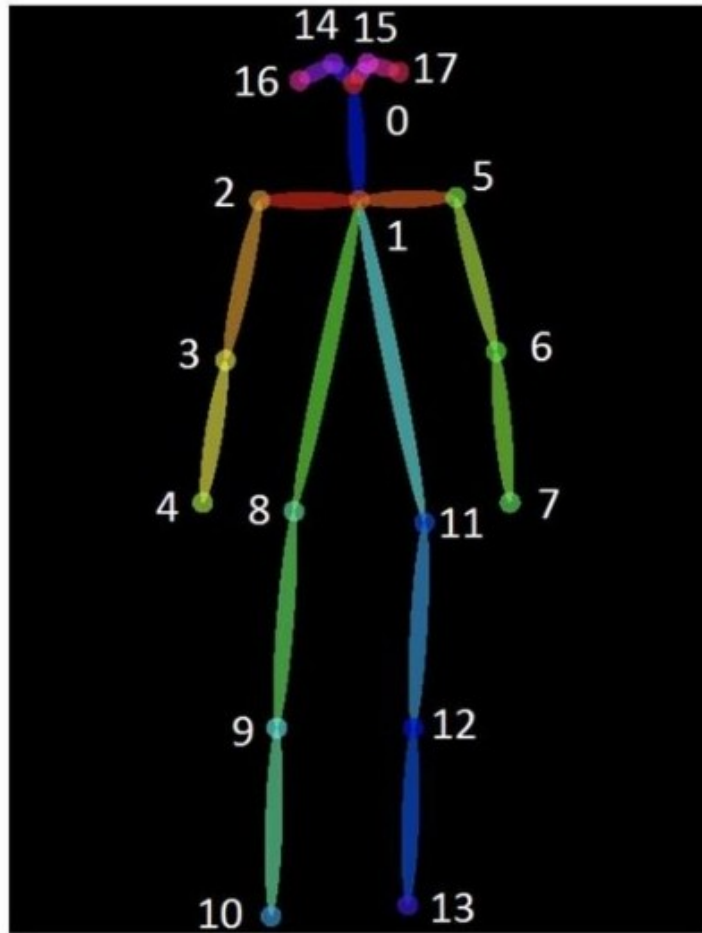
Nr	Name
0	Nose
1	Neck
2	RShoulder
3	RElbow
4	RWrist
5	LShoulder
6	LElbow
7	LWrist
8	RHip
9	RKnee
10	RAnkle
11	LHip
12	LKnee
13	LAnkle
14	REye
15	LEye
16	REar
17	LEar

Human Pose Keypoints



Key points for Emotion Detection

Human Pose Keypoints



Open Pose Library

*Image from MS COCO
Dataset*

Open Pose human pose detection library

Real-time 2D multi-person keypoint detections

- 15, 18, 27-keypoint body/foot keypoint estimation
- 21 hand keypoint estimation
- 70 face keypoint estimation

Real-time 3D single-person keypoint detections

- 3D triangulation with multiple camera views
- Flir camera compatibility

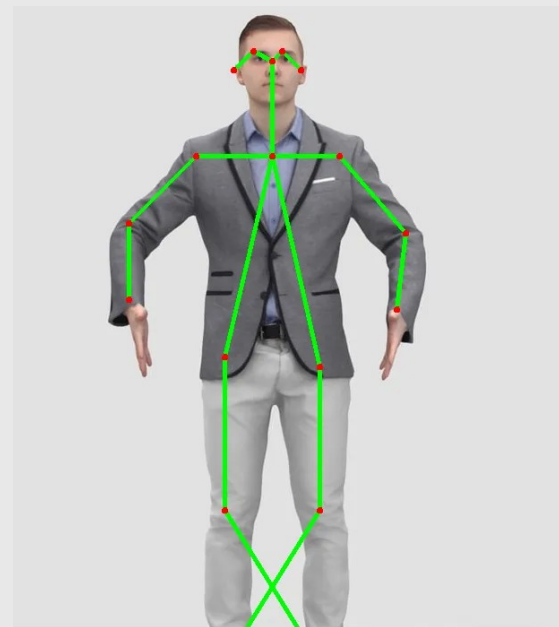
Single-person tracking for speeding up the detection and visual smoothing

Calibration toolbox for the estimation of extrinsic, intrinsic, and distortion camera parameters

license free for non-commercial use.

commercial applications, **non-refundable annual fee of USD 25000**

Open Pose Examples



Some Data sets for human pose detection

- MPII Human Pose Dataset
- 3DPW - 3D Poses in the Wild
- LSPE - Leeds Sports Pose Extended
- DensePose-COCO
- AMASS: Archive of Motion Capture as Surface Shapes
- VGG Human Pose Estimation Datasets
- SURREAL Dataset
- CMU Panoptic Studio Dataset
- **COCO (Common Objects in Context)**
- HumanEva
- Human3.6M
- FLIC (Frames Labelled in Cinema)
- 300-W (300 Faces In-the-Wild)
- Labeled Face Parts in the Wild

Source: <https://encord.com/blog/15-best-free-pose-estimation-datasets/>

Region Convolution Neural Network

R-CNN Fast R-CNN Faster R-CNN

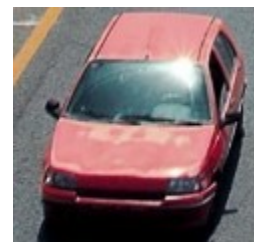
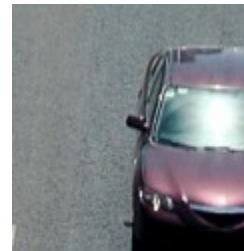
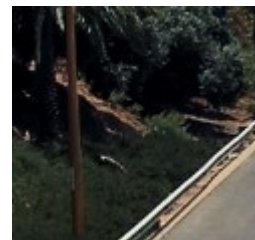
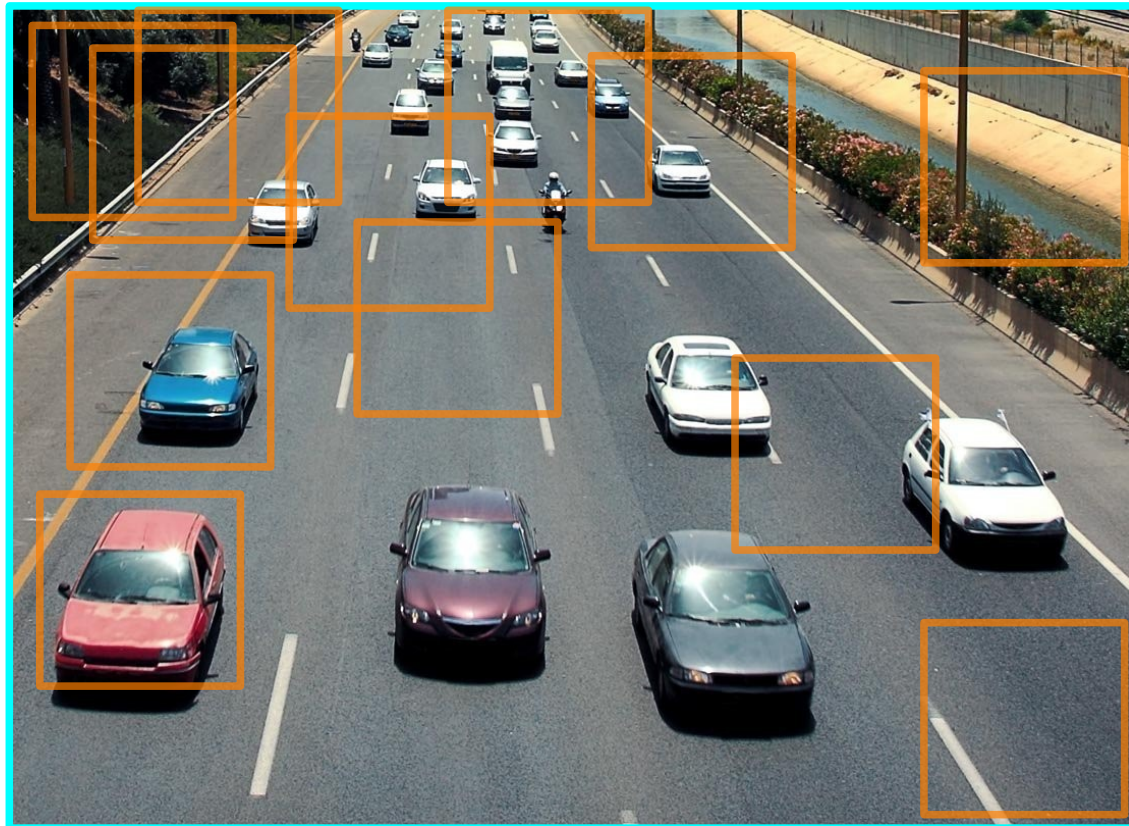
Review of Sliding Window Technique



Input Image

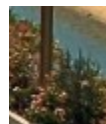
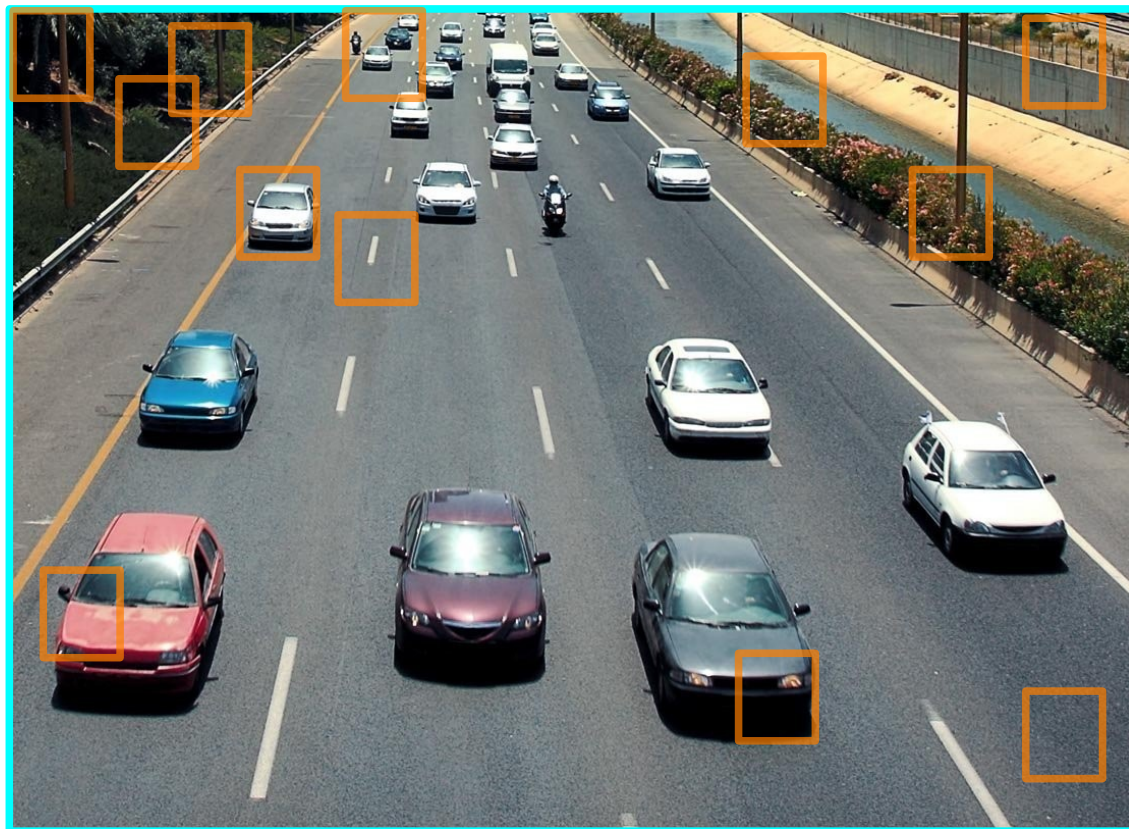
Scan using **MULTIPLE** Windows sizes

Window
of Size 1



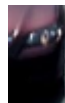
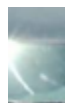
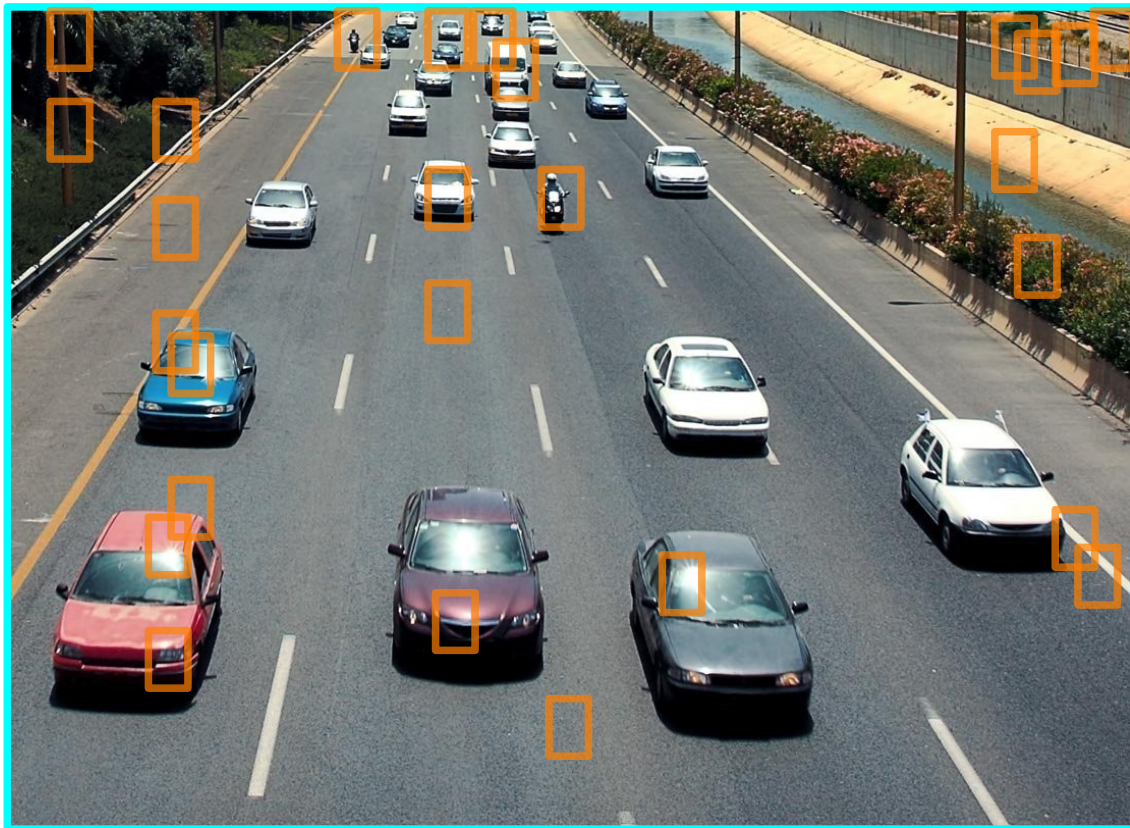


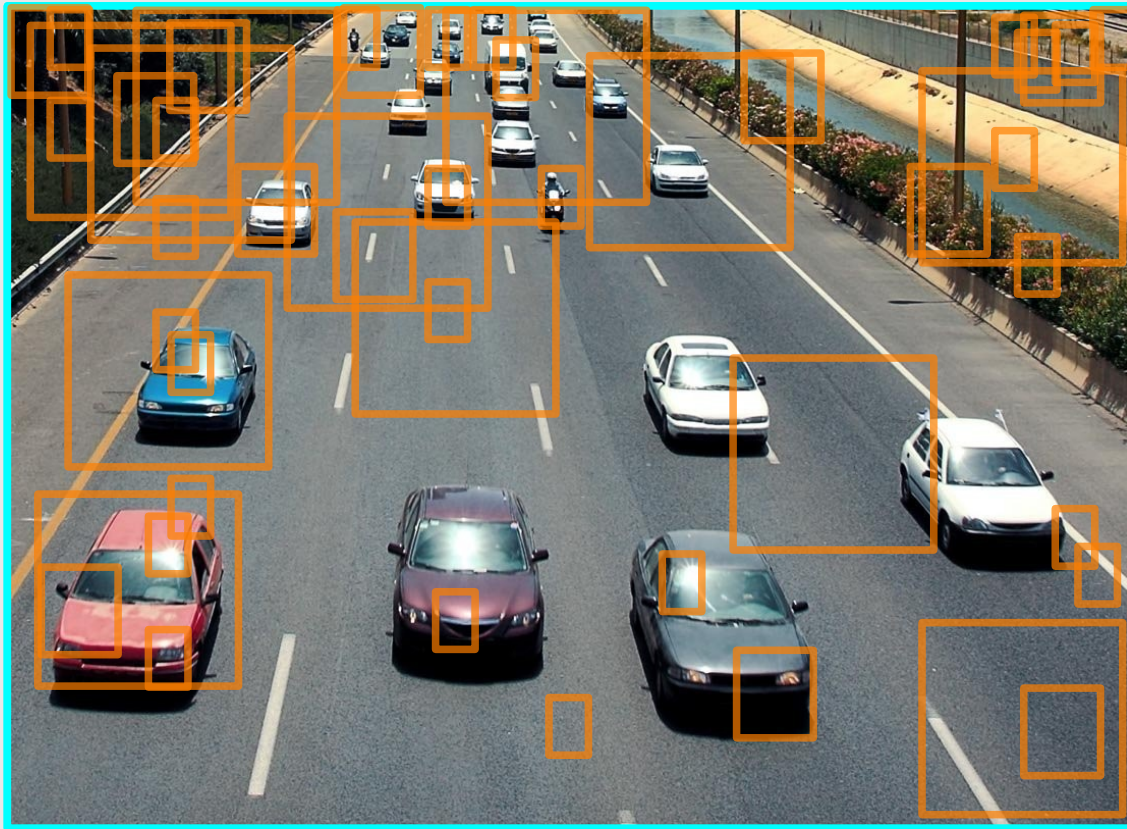
Window
of Size 2





Window
of Size 3





Some of Sliding Window Images

With MULTIPLE Windows sizes



**Classifier
(for detected Classes)**
(SVM, CNN, ...)



**Detected
Class**

Bounding Box is the same size of Sliding Window

Disadvantage of Sliding Window Technique

Main Disadvantage: Very Slow

Reasons for Very Slow Performance:

- **Number of Picked windows is very huge**
- **For CNN classifier, needs to apply convolution to each window content**

Region Based CNN (RCNN)

Family of Techniques
(RCNN – Fast RCNN – Faster RCNN)

Region Based CNN

R-CNN

Fast R-CNN

Faster R-CNN

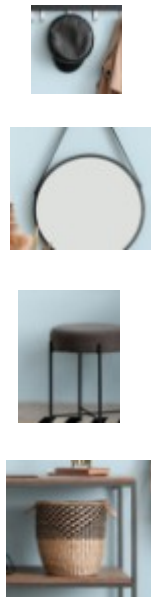
Main Idea

Instead of Huge number of Sliding Window sub-images

Propose some region ONLY for detector



Region Proposals



Object Detector

- Refine Bounding Box
- Object Class

*Bounding Box x, y, W, H
for Proposed Regions*

Up to 2000 proposed Regions

Region Proposal

Region Based CNN

R-CNN (Region proposal + CNN)

There are many techniques for Region Proposals

Most common techniques:

- **Selective Search**
(similarity of pixel colors, intensity, texture, ...)
- **Edge Boxes**
- Each region is extracted from the image then pass through CNN to calculate feature map.
- All regions are clipped to have the same size (in order to have Same feature map size) to be passed to FC for classification and regression of Bounding Boxes.

Region Proposal

[1] Selective Search

Selective Search



Color

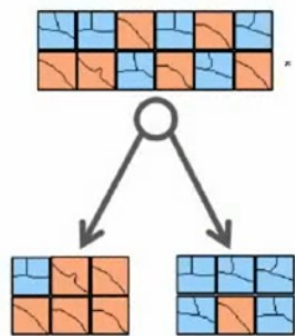
Texture

Circles and
Curves

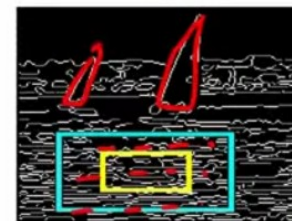
Region Proposal

[2] Edge Boxes

Edge Boxes ✓



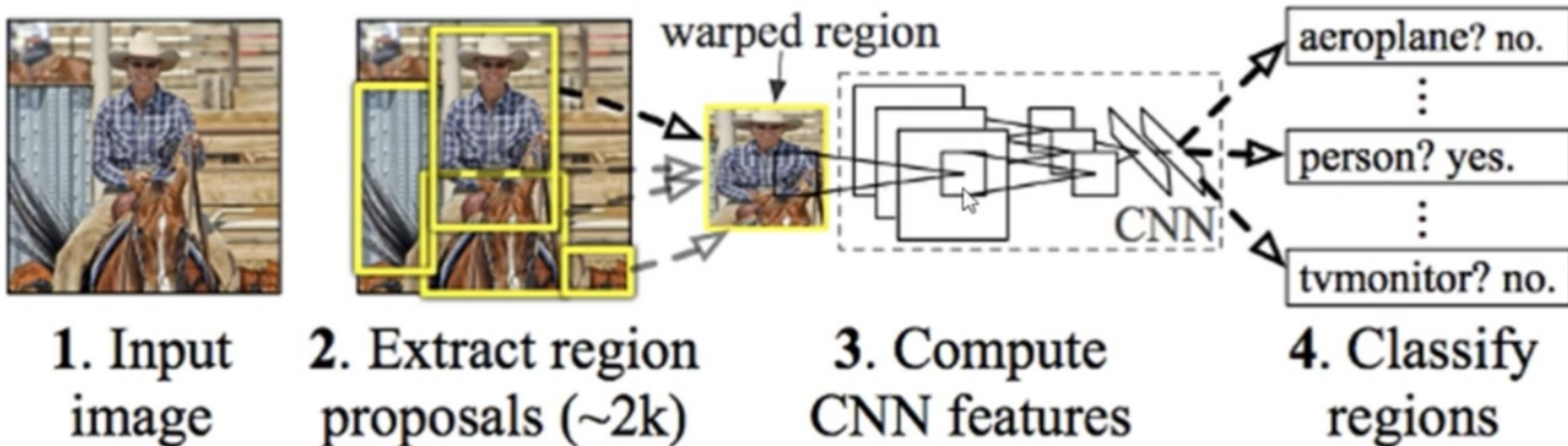
(c)



Region Based CNN

R-CNN (Region proposal + CNN)

R-CNN: *Regions with CNN features*



Region Proposal

Disadvantages Selective Search - Edge Boxes

Main Disadvantage: Very Slow

- 1) Cropping of proposed regions is very slow
- 2) Requires to apply convolution on each proposed image (Multiple Convolutions)
- 3) Same object will be detected in multiple windows (with different Bounding Boxes) due to NOT-Optimized Proposed regions

Solution (to reason #2)

Save time by doing CNN one time only on the whole input images

Fast RCNN

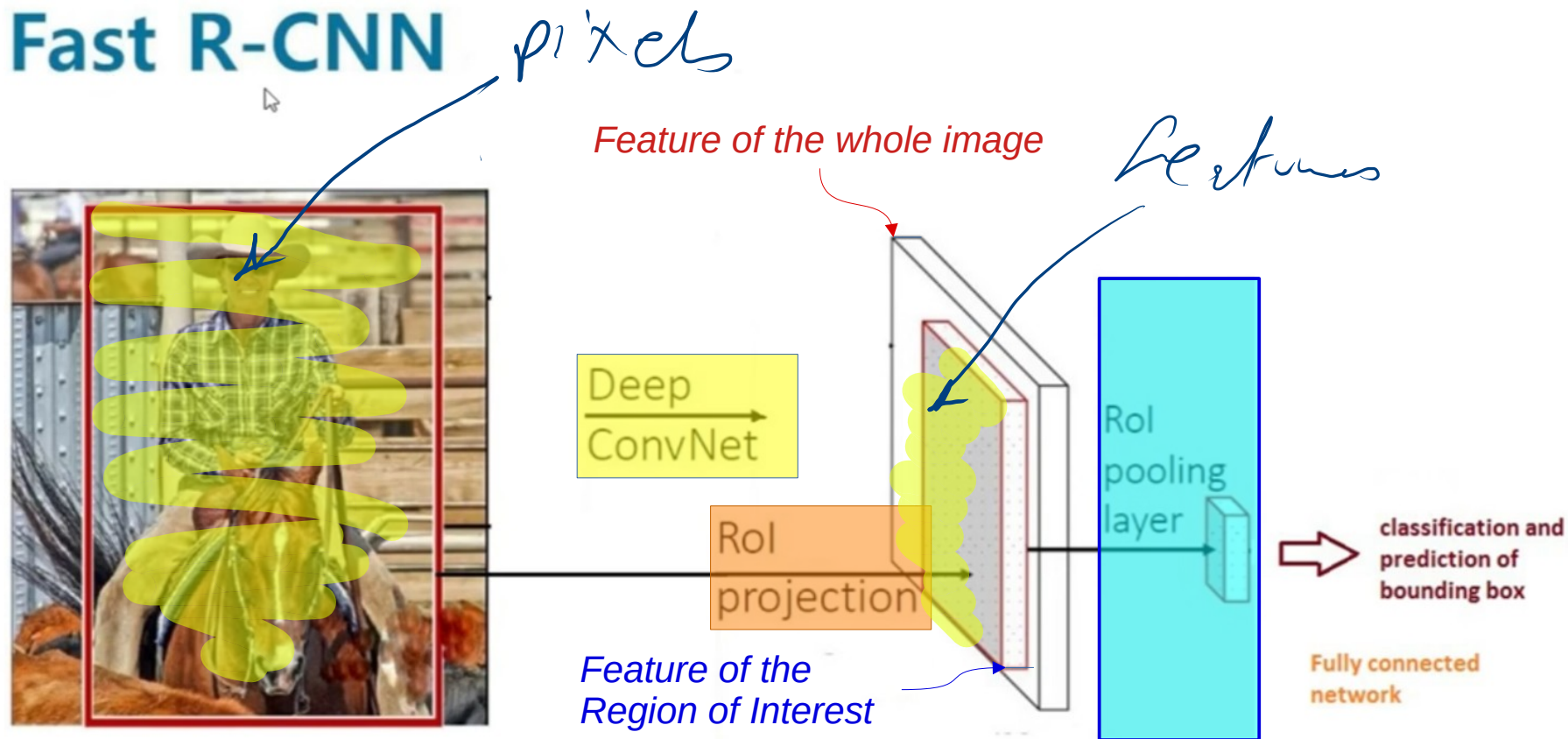
(Extract Regions from Feature Map Domain)

- Region Proposals through **selective search** (same as R-CNN)
The whole image is passed through CNN to get feature map of the whole image
- Each region is extracted from feature map (not from the original image). In other words, Project the bounding box on the Feature map of the image.
- Since regions are of **different sizes**, their corresponding feature map will be of different sizes. Can't be feed to FC NN directly (required to be of same size)
- Instead of cropping regions in RCNN; Apply **ROI pool** layer to fix length of Feature maps of all different-size regions.
- fixed length Feature Maps are passed to FC NN for classification and regression of Bounding Boxes.

Fast RCNN

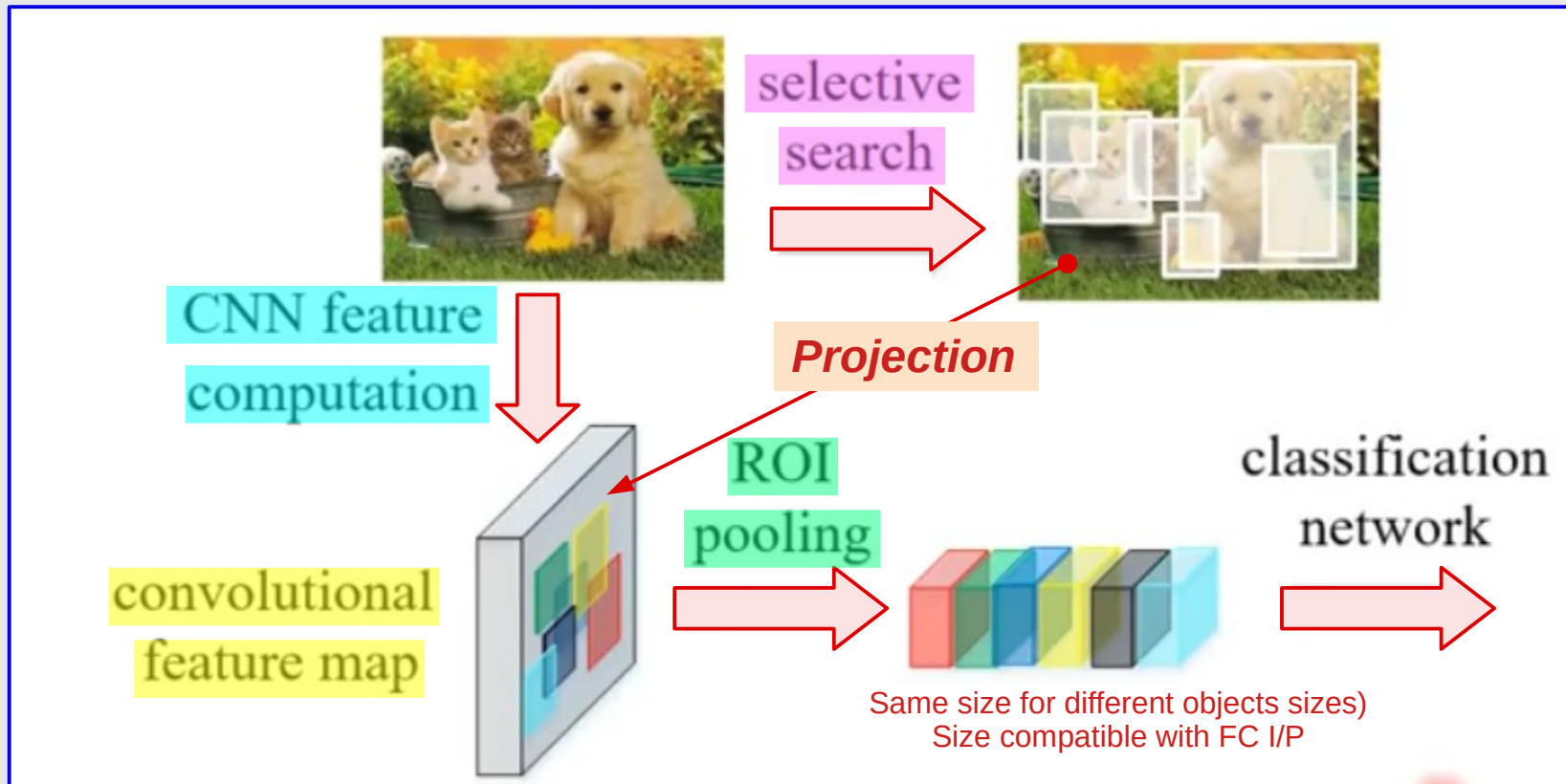
(Extract Regions from Feature Map Domain)

Fast R-CNN



Fast RCNN

(Extract Regions from Feature Map Domain)



Disadvantages of Fast RCNN

- **Slow** (due to using **selective search** technique for region proposal)
- Still **large number of Proposed regions**
- Same object will be detected in multiple windows (with different Bounding Boxes)

Solution

- **Automate Region Proposal using Trained Network** (instead of handcrafted techniques such as Selective Search and Edge Boxes)
- **Optimize** number of Proposed Regions

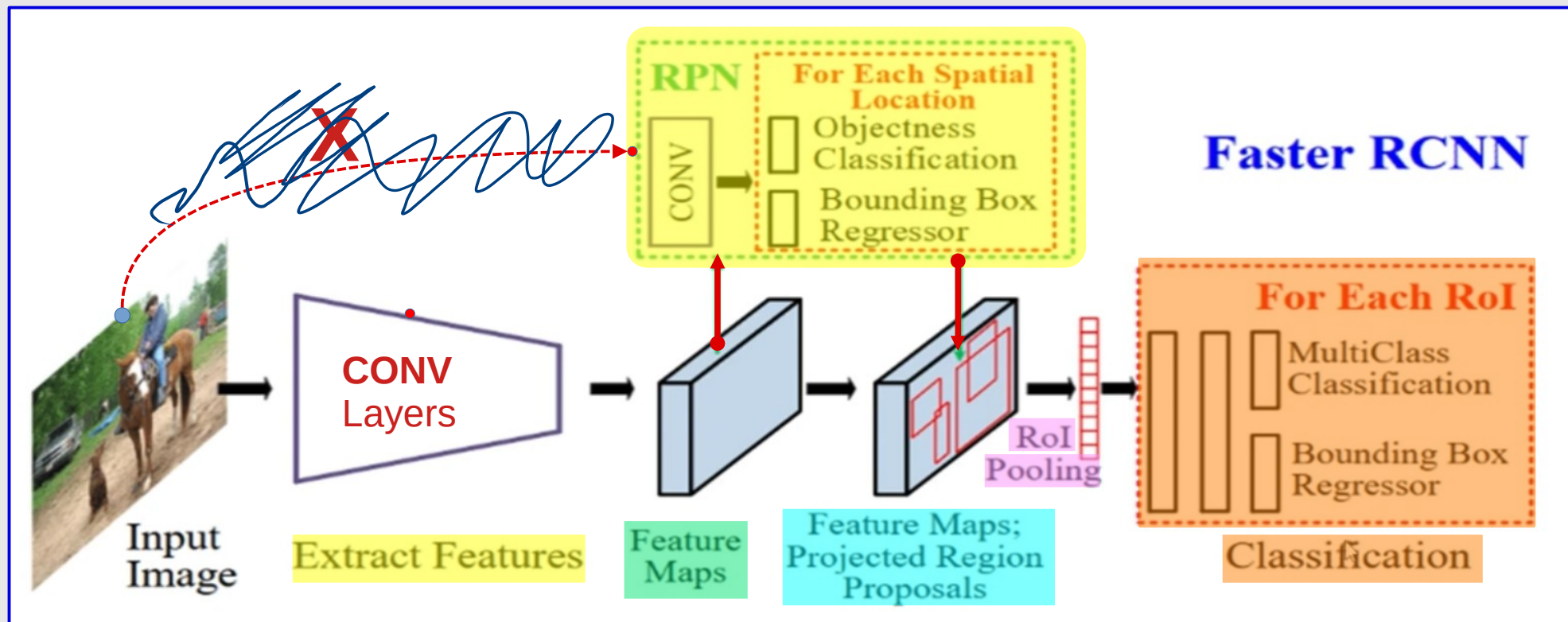
Faster RCNN

(Region proposal Network (RPN) + CNN)

- Region Proposals through *the trained “Region Proposal Network RPN”*
The whole image is passed through CNN to get feature map of the whole image
- Each region is extracted from feature map (not from the original image). In other words, Project the bounding box on the Feature map of the image.
- Since regions are of **different sizes**, their corresponding feature map will be of different sizes. Can't be feed to FC NN directly (required to be of same size)
- Instead of cropping regions in RCNN; Apply **ROI pool** layer to fix length of Feature maps of all different-size regions.
- fixed length Feature Maps are passed to FC NN for classification and regression of Bounding Boxes.

Faster RCNN

(Region proposal Network (RPN) + CNN)



[1] Region Proposed Network

- Architecture of Region Proposed Network (RPN)
- How to prepare Training Data for RPN
- How to Train RPN Network

[2] How ROI Pooling is achieved

[3] Detector Network

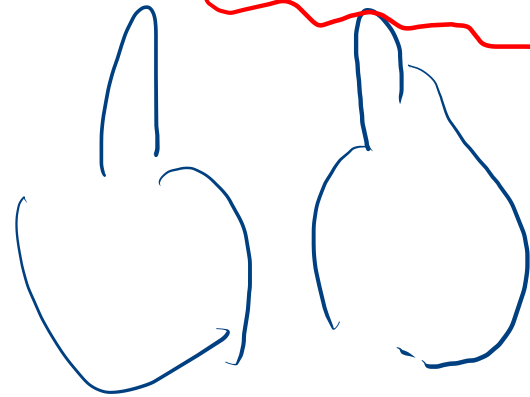
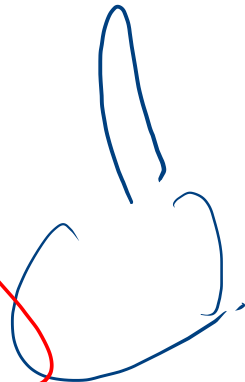
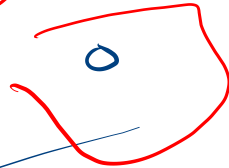
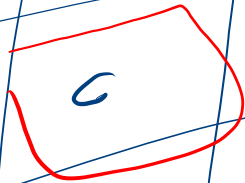
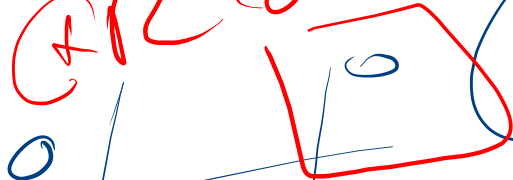
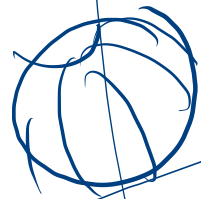
- Architecture of Detector Network (FC)
- How to prepare Training Data for RPN and Detector Network
- How to Train Detector Network (Loss Function)

[4] How to Use Faster RCNN in Segmentation Mask-RCNN

Course project

~~7-May~~

* blue
* red



2 students

video

Red

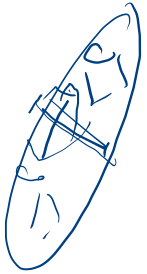
4
blue

Bonus Project (optional) 5%

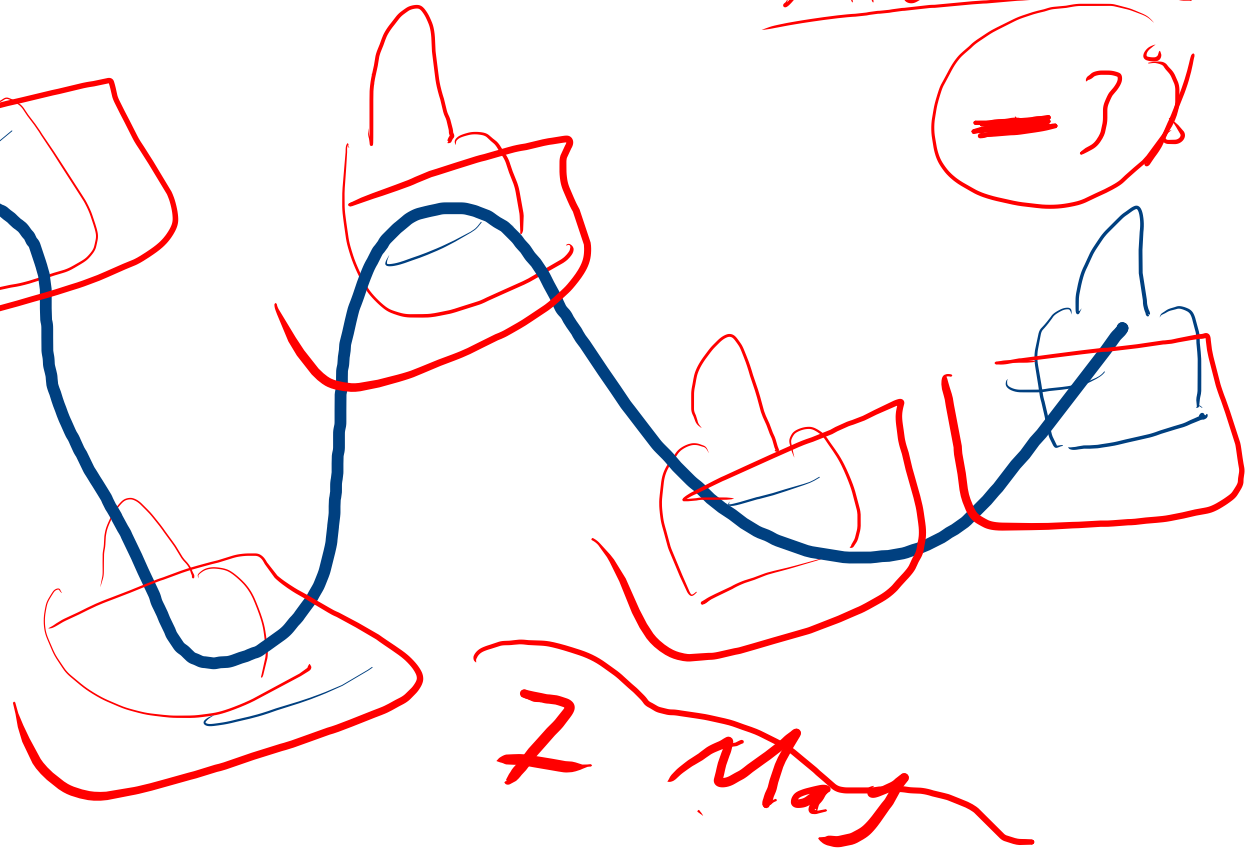
add to
Midterm

Mobile APP

GP line



Camera
then
Mobile APP



-3%